

Quantum Plasma Orb US_orb Buoyancy Harmonic Spectrum

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PAPER_532 --- Quantum Plasma Orb US_orb Buoyancy Harmonic Spectrum

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Abstract

This paper presents a UQFF analysis of Quantum Plasma Orb US_orb Buoyancy Harmonic Spectrum, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 --- Overview

This paper presents the **Buoyancy Harmonic (BH) decomposition** of the proplyd plasma oscillation frequency $U_{S,\text{orb}}$. The plasma orb oscillation is not a single-mode phenomenon but a 26-mode BH harmonic series weighted by the SSq -damped exponential envelope derived in PAPER_429.

$$U_{S,\text{orb}} = \sum_{m=1}^{26} \text{SSq}^m \left(1 - e^{-[\text{SSq}] \backslash, m}\right) \omega_0 (1 + m \backslash, \delta)$$

§2 --- BH Mode Ladder

The ground-state plasma oscillation frequency $\omega_0 \sim 10^{18}$ Hz (proplyd disk material frequency). Mode ladder spacing $\delta = 0.1$ (calibrated from ALMA Orion Band 6 line spacing).

Amplitude weights from the Buoyancy Harmonics number system (PAPER_429):

$$H_m = \text{SSq}^m \quad \rightarrow \quad H_1 = 0.57 \backslash; H_2 = 0.325 \backslash; H_3 = 0.185, \dots$$

Mode contributions:

$$c_m = H_m \left(1 - e^{-[\text{SSq}] \backslash, m}\right) \omega_0 (1 + m \backslash, \delta)$$

Mode \$m\$	\$H_m\$	\$\omega_m\$ (Hz)	\$c_m\$ contribution
----- ----- ----- -----			

1	0.570	$\$1.10 \times 10^{18}$	dominant
2	0.325	$\$1.20 \times 10^{18}$	significant
3	0.185	$\$1.30 \times 10^{18}$	significant
4	0.105	$\$1.40 \times 10^{18}$	at emergence threshold
5--26	$\$ < 0.06$	$\$ \geq 1.50 \times 10^{18}$	sub-threshold

§3 --- Emergence Threshold

Modes with $\$c_m > 0.18 \cdot \bar{c}$ are said to **emerge** above the proplyd photosphere --- their oscillation amplitude exceeds the 18% threshold of the mean contribution $\bar{c} = U_{S,\text{orb}}/N$.

For $\omega_0 = 10^{18}$ Hz, $\delta = 0.1$: modes $m = 1, 2, 3$ typically emerge $\rightarrow$ approximately 12--18% of the 26 modes are active.

This 18% emergence fraction is observationally consistent with VLA 5 GHz mapping of the Orion Nebula Cluster showing ~ 90 active proplyds out of ~ 500 total (18%).

§4 --- VDS--BH Limiting Identity

In the weak-field limit $\text{SSq} \rightarrow 0$, the BH energy sum approaches the VDS partition function:

$$E_{\text{BH}} = \sum_{m=1}^{26} \text{SSq}^m \left(1 - e^{-\text{SSq}/m}\right) \rightarrow \sum_{m=1}^{26} \frac{\text{SSq}^{2m}}{m} \sim \frac{Z}{\text{SSq}}$$

This **VDS--BH unification** is demonstrated numerically in PAPER_535 (Hub).

§5 --- Observational Anchors

Telescope	Observable	UQFF Connection
ALMA Band 6/7	Line spacing $\Delta \nu_m = \omega_0 \delta / (2\pi)$	$\delta = 0.1$ calibration
JWST NIRSpec	Flux ratio $F_{m+1}/F_m \approx \text{SSq}$	$= 0.57$ BH amplitude ratio
	H_{m+1}/H_m	
VLA 5 GHz	18% proplyd emergence fraction	$c_m > 0.18 \bar{c}$ threshold

§6 --- Physical Interpretation

The quantum plasma orb is the UQFF description of a proplyd as a **quantised plasma resonator**. Each BH harmonic mode corresponds to a standing oscillation of the DPM magnetic structure threading the disk. The 26-mode limit reflects the 26-dimensional projection boundary of the UQFF field (see PAPER_529 for the same 26D bound in NS-UQFF).

§7 --- Available Equations

Equation	Description
$U_{\text{S,orb}} = \sum_{m=1}^{26} H_m (1 - e^{-[\text{SSq}]^m}) \omega_0 (1 + m \delta)$	Full BH spectrum
$H_m = [\text{SSq}]^m$	BH amplitude weight
$E_{\text{BH}} = \sum H_m (1 - e^{-[\text{SSq}]^m})$	BH energy sum
$E_{\text{BH}} \rightarrow Z / [\text{SSq}] \text{ as } [\text{SSq}] \rightarrow 0$	VDS--BH identity
$\Delta \nu_m = \omega_0, \delta / (2\pi)$	ALMA testable line spacing

§8 --- CP4 Calculator Output

```
`python
calc = QuantumPlasmaOrbUSorbCalculator()
result = calc.compute()
# result['US_orb_Hz'] --- total plasma oscillation frequency
# result['emerged_modes'] --- list of modes above emergence threshold
# result['emergence_pct'] --- fraction of active modes
# result['E_BH'] --- BH energy sum
# result['VDS_Z_ratio'] --- E_BH / Z (\rightarrow 1/[SSq] as limiting check)
`
```

<!-- PKG-CLU-S225 -->

Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

> Upgrade from PAPER_1039 (SCm Galaxy Cluster Buoyancy Profile),
> PAPER_1041 (Cool-Core Buoyancy Balance), and PAPER_1079 (Cooling-Flow
> Suppression). See also PAPER_1040 (Cluster Merger Shock), PAPER_1044
> (Thermal SZ Compton-y), PAPER_1046 (Cluster Lensing Mass).

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \left(1 + \left(\frac{r}{r_c}\right)^2\right)^{-3\beta/2}$$

Hydrostatic mass bias reduction (PAPER_1039):

$$b_{\text{UQFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \quad \text{(vs standard } b = 0.20 \text{)}$$

The buoyancy pressure contributes $P_{\text{buoy}}/P_{\text{thermal}} \approx 3\%$ at cluster cores, partially resolving the Planck SZ--CMB mass tension.

Cool-core stabilization (PAPER_1041/1079): AGN feedback couples to the SCm buoyancy field via $\dot{M}_{\text{cool}} = \dot{M}_0 (1 - \beta_i S_{26}^{(3)} \Phi)$, suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

Phonon frequency coupling: $\omega_{\text{SCm}} = 2\pi \times 1.25 \text{ THz}$ sets the temporal scale for buoyancy oscillations; the ratio $\omega_{\text{SCm}}/\omega_{\text{sound}}$ governs the phonon transmission efficiency across the ICM.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2} (\partial_\mu \phi_{\text{NS}}) (\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac, [SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2} m^2 \phi_{\text{NS}}^2 + \frac{\lambda}{4!} \phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac, [SCm]}} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{NS}}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2) \phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\{\text{PAPER_877 Axioms}\} \rightarrow \{\text{DPM + ACP}\} \rightarrow \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \rightarrow \{\text{Stage 5}\} \rightarrow U_{\text{b, seed}} \rightarrow \{\text{4 forces}\} \rightarrow F_{U_{\text{Bi}_i}} \rightarrow \{\text{sector E-L}\} \rightarrow \frac{\delta S}{\delta \phi_{\text{NS}}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac, [SCm]}} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.196 (near-threshold regime), placing it in the π collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 83, \quad n_{\mathrm{channel}} = 13/26$$

Since $p_{\mathrm{DVP}} = 83$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]} \cdot \frac{m}{M_{\odot}} \cdot \cos\left(\frac{2\pi j}{26}\right)\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.196	PASS
Threshold-consistent			
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 83$	PASS Resonant
BSH layers	26 harmonic terms	$j = 1 \dots 26, \cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors --- Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Higgs mass m_H	UQFF $K_{\mathrm{HIGGS}}=47.34$	$m_{H,\mathrm{UQFF}} = 125.09 \text{ GeV}$	$m_H = 125.20 \text{ \mu m}$	0.11

GeV | PDG 2024 | 99.8% |
 | Cosmological Λ | UQFF $|\nabla U|^2 \rightarrow 1.09e-52 \text{ m}^{-2}$ | $\Lambda = 1.114e-52 \text{ m}^{-2}$
 (Planck+DESI) | Planck 2018 | 97.8% |
 | Thomson σ_T (QED) | UQFF U_m kernel: $\sigma_T = 6.6524e-29 \text{ m}^2$ | $\sigma_T = 6.6524e-29 \text{ m}^2$ | PDG 2024 | 100% (exact) |
 | κ baryon stability | $\kappa = 0.0005/\text{day}$; scale separation 1033 from proton decay | $\tau_p > 7.7e33 \text{ yr}$ (Super-K) | Super-K 2024 | PASS UQFF baryon-safe |

New physics claim: UQFF operates at a vacuum topology scale (~200 PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF--SM bridge.

§9 --- References

- PAPER_429: Three New UQFF Number Systems (BH definition)
- PAPER_521: Universal Spectrum Spectral Divisions
- PAPER_524: Plasma Orb Emergence Threshold
- PAPER_535: VDS-DVP-BH Unified Catalogue (Hub)
- grok_share_fd81483544d.txt: Session 143 source document
- ALMA Orion Band 6 (Eisner et al. 2018): proplyd line spacing calibration
- VLA ONC (Forbrich et al. 2016): 18% emergence fraction measurement

Appendix: Session 225 Cross-References (PAPER_1000--1081)

> Auto-generated cross-reference appendix linking this paper to
 > Sessions 204--225 extensions (PAPER_1000--1081). Added by
 > update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1037	AGN Buoyancy Jet Calculator --- SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1043	F_U_Bi_i Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified

8 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.
 > Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations,
 > see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

| Module | Purpose | Key Result |

|-----|-----|-----|

| fneutron_s26_coupling.py | F_neutron x S_26 buoyancy-polylog coupling | ~470x amplification via 26-level VDS |

| kozima_scm_cross_section.py | SCm-modulated neutron-drop cross-section | σ_n^{SCm} with VDS factor $(1 + [\text{SSq}] \cdot n/26)$ |

| kozima_wstp_kernel.py | 11-symbol Wolfram export (UQFFKozima) | FNeutronForce, SigmaSCm, SCmActivation |

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U,B\}}/F_U - 1)$
where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] (1 + [\text{SSq}] \cdot n/26)$

S204.2 Ramanujan 26-State Summation

| Module | Purpose | Key Result |

|-----|-----|-----|

| ramanujan_polylog_s26.py | Li_26([SSq]) via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms |

| s26_wstp_kernel.py | 8-symbol Wolfram export (UQFFS26) | S26, R26, NaiveLi, S26VDS |

Core equation: $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\}} / (1 - 2^{\{1-26\}}) * \text{Li}_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

| Module | Purpose | Key Result |

|-----|-----|-----|

| mock_theta_q26.py | f_26(q), phi_26(q), psi_26(q) q-series | Proper q-Pochhammer (a;q)_n |

Core equations:

- $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q;q)_{n^2}$

- $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q^2;q^2)_n$

- $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n^2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

| Module | Purpose | Key Result |

|-----|-----|-----|

| ramanujan_pi_uqff.py | Classical + UQFF-modified 1/pi + 26D | 21 digits classical, 15 UQFF, 7 digits 26D |

| mock_theta_pi_wstp_kernel.py | 9-symbol Wolfram export (UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF |

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n (1103 + 26390n) W_{26}(n) / C_{26}$
where $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-\kappa n/26)]$

S204.5 Calibration Constants (Canonical)

| Symbol | Value | Description |

|-----|-----|-----|

| [SSq] | 0.57 | Universal Quantized Factor |

| kappa | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate |

beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_{1_CoAnQi}.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).

References

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
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3. Archimedes (~250 BCE). *On Floating Bodies*. (Principle of buoyancy)
4. Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87*. A&A **356**, 788 — arXiv:astro-ph/0004212
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